

Test Verification for PCM-071 (phyCORE-AM62x): Low Power Mode Testing

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1. Recommended Changes Table

Test Run	Test Case Execution	SCH	LAY	BoM	SW	RWK	MEC	Recommended Changes	Products to Change
Low Power Mode Testing	SOM Current Consumption in Low-Power Modes (2.2.1 Test Case Execution 1)				X			Update firmware to completely turn off MCU to enable multiple uses of Suspend-2-RAM.	PL1573.1

2. Low Power Mode Testing

This testing captures the following:

- Current and power consumption in Linux
- Current and power consumption in Suspend-2-RAM low-power mode
- Current and power consumption in MCU Only low-power mode

2.1. Test Objectives

Test Case 1 (SOM Current Consumption in Low-Power Modes):

This test captures the readings for the typical current consumption of the SOM during operation in Low-Power Modes

2.2. Test Results

2.2.1 Test Case Execution 1 (SOM Current Consumption in Low-Power Modes)

[executed by cbrown on 2026-08-12]

Script:

Step: Replace R492 on the carrier board with a power shunt resistor

Results: For this testing, a 70 mOhm resistor was populated.

Step: For each power mode, measure the current and power consumption of the system.

Results: This testing was performed using the PD25.1.1 BSP, which enables the BTN1 as a wakeup signal from Suspend-2-RAM low-power mode.

This worked reliably 1 time per boot, but locked up when entering low-power mode a second time.

It seems that the MCU firmware has to stop all the way before the system can resume from sleep. It is possible that the firmware does not start up correctly after the first resume, and then it just hangs when you try to do the second sleep. This would seem to explain both the higher power usage and the inability to resume the second time. To correct this, we found the correct remoteproc for the m5 system:

```
root@phyboard-lyra-am62xx-3:~# cat /sys/class/remoteproc/remoteproc0/name
5000000.m4fss
```

```
root@phyboard-lyra-am62xx-3:~# cat /sys/class/remoteproc/remoteproc1/name
78000000.r5f
```

Test Verification for PCM071: Low Power Mode Testing

In this case, remoteproc0. Then disabled the m4 firmware:

```
root@phyboard-lyra-am62xx-3:~# echo stop >
/sys/class/remoteproc/remoteproc0/state
[ 153.890192] remoteproc remoteproc0: stopped remote processor
5000000.m4fss
```

After this, we were able to enter and leave the Suspend-2-RAM low-power mode as many times as desired.

```
root@phyboard-lyra-am62xx-3:~# echo mem > /sys/power/state
[ 163.995928] PM: suspend entry (deep)
[ 164.000262] Filesystems sync: 0.000 seconds
[ 164.011016] Freezing user space processes
[ 164.017127] Freezing user space processes completed (elapsed 0.001
seconds)
[ 164.024310] OOM killer disabled.
[ 164.027640] Freezing remaining freezable tasks
[ 164.033948] Freezing remaining freezable tasks completed (elapsed 0.001
seconds)
[ 164.041454] printk: Suspending console(s) (use no_console_suspend to
debug)
[ 164.075010] Disabling non-boot CPUs ...
[ 164.077456] psci: CPU1 killed (polled 0 ms)
[ 164.080172] Enabling non-boot CPUs ...
[ 164.080604] Detected VIPT I-cache on CPU1
[ 164.080659] GICv3: CPU1: found redistributor 1 region
0:0x00000000018a0000
[ 164.080724] CPU1: Booted secondary processor 0x0000000001 [0x410fd034]
[ 164.082107] CPU1 is up
[ 164.082979] ti-sci 44043000.system-controller: ti_sci: wakeup source:0x80,
pin:0x75, mode:0x0
[ 164.104183] am65-cpsw-nuss 8000000.ethernet: set new flow-id-base 19
[ 164.113447] am65-cpsw-nuss 8000000.ethernet end0: PHY [8000f00.mdio:01]
driver [TI DP83867] (irq=POLL)
[ 164.113474] am65-cpsw-nuss 8000000.ethernet end0: configuring for
phy/rgmii-rxid link mode
[ 164.121318] am65-cpsw-nuss 8000000.ethernet end1: PHY [8000f00.mdio:03]
driver [TI DP83867] (irq=POLL)
[ 164.121335] am65-cpsw-nuss 8000000.ethernet end1: configuring for
phy/rgmii-rxid link mode
[ 164.212205] OOM killer enabled.
[ 164.215356] Restarting tasks ... done.
[ 164.225079] random: crng reseeded on system resumption
[ 164.232937] PM: suspend exit
root@phyboard-lyra-am62xx-3:~# cat /proc/interrupts | grep WAKEUPGPIO
455: 1 0 GPIO 23 Edge -davinci_gpio WAKEUPGPIO
484: 0 0 pinctrl 468 Edge WAKEUPGPIO:wakeup
root@phyboard-lyra-am62xx-3:~# echo mem > /sys/power/state
[ 181.023750] PM: suspend entry (deep)
[ 181.027985] Filesystems sync: 0.000 seconds
```

Test Verification for PCM071: Low Power Mode Testing

```
[ 181.035775] Freezing user space processes
[ 181.042147] Freezing user space processes completed (elapsed 0.002
seconds)
[ 181.049184] OOM killer disabled.
[ 181.052439] Freezing remaining freezable tasks
[ 181.058348] Freezing remaining freezable tasks completed (elapsed 0.001
seconds)
[ 181.065849] printk: Suspending console(s) (use no_console_suspend to
debug)
[ 181.098449] Disabling non-boot CPUs ...
[ 181.100987] psci: CPU1 killed (polled 0 ms)
[ 181.102417] Enabling non-boot CPUs ...
[ 181.102857] Detected VIPT I-cache on CPU1
[ 181.102914] GICv3: CPU1: found redistributor 1 region
0:0x000000000018a0000
[ 181.102983] CPU1: Booted secondary processor 0x0000000001 [0x410fd034]
[ 181.104470] CPU1 is up
[ 181.105216] ti-sci 44043000.system-controller: ti_sci: wakeup source:0x80,
pin:0x75, mode:0x0
[ 181.125469] am65-cpsw-nuss 8000000.ethernet: set new flow-id-base 19
[ 181.134552] am65-cpsw-nuss 8000000.ethernet end0: PHY [8000f00.mdio:01]
driver [TI DP83867] (irq=POLL)
[ 181.134580] am65-cpsw-nuss 8000000.ethernet end0: configuring for
phy/rgmii-rxid link mode
[ 181.141357] am65-cpsw-nuss 8000000.ethernet end1: PHY [8000f00.mdio:03]
driver [TI DP83867] (irq=POLL)
[ 181.141370] am65-cpsw-nuss 8000000.ethernet end1: configuring for
phy/rgmii-rxid link mode
[ 181.228836] OOM killer enabled.
[ 181.231982] Restarting tasks ... done.
[ 181.239596] random: crng reseeded on system resumption
[ 181.245433] PM: suspend exit
```

The results of power testing in low-power modes are below:

Case	Current (mA)	Power (mW)	Difference
Idle in Linux, not in Low-Power Mode	297.1	1,485.7	0%
Suspend-2-RAM Low-Power Mode	25.7	128.6	-91.3%
MCU-Only Low-Power Mode	N/A	N/A	N/A

The MCU/M4 demo firmware deployed to our current BSP release does not work in combination with the MCU-only mode. It may be available in future versions.

2.3. Test Summary

Test Name	Test Status	Changes	Recommended Changes	Products to Change
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Test Verification for PCM071: Low Power Mode Testing

Test Case Execution 1 (SOM Current Consumption in Low-Power Modes)	Pass - Change Required	Software	Update firmware to completely turn off MCU to enable multiple uses of Suspend-2-RAM.	PL1573.1
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